Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Authentication	USN-1	As a developer, I can register for an API key by providing name, email, and password.	2	High	Dev 1
Sprint-1	Authentication	USN-2	As a developer, I will receive a verification email to activate my API key access.	1	High	Dev 1
Sprint-1	Authentication	USN-3	As a developer, I can login using OAuth 2.0 (Google/GitHub) for instant access.	2	Medium	Dev 2
Sprint-1	Authentication	USN-4	As a system, I can authenticate incoming API requests using Bearer Tokens and API Keys.	2	High	Dev 2
Sprint-1	Weather Service	USN-5	As a user, I can fetch current weather data by passing latitude and longitude parameters.	1	High	Dev 3
Sprint-2	AI Forecasting	USN-6	As a user, I can query AI-predicted 7-day precipitation forecasts for a given city.	3	High	Dev 3
Sprint-2	AI Forecasting	USN-7	As a user, I can request severe weather alert predictions generated by the ML model.	5	High	Dev 4
Sprint-2	Dashboard	USN-8	As an admin, I can monitor real-time API request analytics and throughput metrics.	3	Medium	Dev 2
Sprint-3	Optimization	USN-9	As a developer, I want cached weather queries via Redis to reduce API latency below 100ms.	3	Medium	Dev 1

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	25 Sep 2026	30 Sep 2026	20	30 Sep 2026
Sprint-2	20	6 Days	01 Oct 2026	06 Oct 2026	18	06 Oct 2026
Sprint-3	20	6 Days	07 Oct 2026	12 Oct 2026	20	12 Oct 2026
Sprint-4	20	6 Days	13 Oct 2026	18 Oct 2026	20	18 Oct 2026
Sprint-5	15	5 Days	19 Oct 2026	23 Oct 2026	15	23 Oct 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day):

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$